

A Multi-Agent, Local-First Edge AI Architecture for Autonomous Residential Energy Optimisation Under Dynamic Tariffs

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Abstract

Residential energy systems remain inefficient despite the proliferation of smart home technologies, largely due to limitations in existing Home Energy Management Systems (HEMS). These limitations include heavy cloud dependence, constrained learning capabilities, and poor integration with dynamic energy pricing structures. This paper presents a multi-agent, local-first edge AI architecture for autonomous residential energy optimisation under dynamic tariffs, supported by empirical validation on real-world data and a deployable hardware prototype. The system decomposes intelligence into three cooperative agents—(i) an Energy Agent for tariff-aware load scheduling using gradient-boosted time-series forecasting, (ii) an Automation Agent that models occupant behaviour using reinforcement learning and probabilistic inference, and (iii) a System Agent that ensures long-term robustness—augmented by a neuromorphic Neural Autonomous Residential Engine (NARE) that learns occupant habits via spike-timing-dependent plasticity (STDP). We train and validate the forecasting pipeline on **54,197 real half-hourly smart-meter records** from the Low Carbon London (LCL) dataset across 20 UK households, achieving $R^2 = 0.89$ (MAE = 0.10 kWh) on held-out test data with a dual-head quantile regression architecture for demand-spike prediction. A greedy scheduler applied to these real load profiles yields a **26.4% mean cost reduction**, equivalent to **£476/year** per household under simulated dynamic tariffs. The NARE brain demonstrates autonomous synaptic potentiation in a 30-event teaching scenario, with sensor-to-actuator synapse weights growing from 0.05 to 0.15 without labelled training data. All software runs on an NVIDIA Jetson Orin Nano housed in a custom 6061-T6 aluminium enclosure (hub: £600–700; typical four-bedroom installation: £2,000–2,500 with professional fitting). Complementary digital-twin Monte Carlo simulations across four tariff scenarios confirm system stability under extreme volatility. These results demonstrate a tangible, reproducible proof-of-concept that bridges theoretical energy optimisation and commercially viable residential deployment.

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1 Introduction

1.1 Background and Motivation

Residential buildings account for a substantial share of global energy consumption, with heating, cooling, and appliance usage constituting the primary load components. Despite the increasing availability of smart home technologies, most residential environments remain reactive rather than intelligent, relying heavily on fixed schedules, manual control, or simple rule-based automation. As electricity markets transition toward dynamic pricing models, including time-of-use (ToU) and real-time tariffs, the economic and operational limitations of static control strategies become increasingly evident.

Current smart home platforms are predominantly cloud-centric, mediating sensing, decision-making, and control through remote infrastructure. While this paradigm enables scalable data aggregation, it introduces several critical vulnerabilities: (i) privacy concerns arising from the continuous transmission of behavioural data; (ii) latency constraints that hinder real-time control applications; and (iii) a strict reliance on network connectivity, which diminishes overall system robustness. These limitations are particularly acute in residential environments, where reliability, responsiveness, and user trust are paramount [1].

Simultaneously, advances in machine learning (ML), particularly reinforcement learning (RL) and time-series forecasting, have demonstrated strong potential for demand-side energy optimisation. Prior studies report energy savings in the range of 10% to 30% utilizing learning-based approaches [2, 3]. However, these methods are typically evaluated in highly controlled environments, assume the presence of extensive sensing infrastructure, or fail to account for real-world deployment constraints such as hardware limitations, occupant variability, and long-term system maintenance.

1.2 Research Gap

Despite significant progress across academia and industry, a persistent gap remains between theoretically optimal energy management approaches and systems that can be practically deployed in residential settings. Existing solutions largely fail to simultaneously address the following challenges:

- **Local-First Intelligence:** Most systems rely on cloud-based computation, with limited exploration of fully on-device learning and control.
- **Cold-Start and User Acceptance:** Learning-based systems often exhibit poor initial performance, leading to reduced user trust and subsequent disengagement.
- **Scalable Sensing:** Appliance-level monitoring typically requires intrusive hardware, fundamentally limiting scalability and broad adoption.
- **Integrated Optimisation:** Energy cost, occupant comfort, and system constraints are rarely jointly optimised within a unified framework.
- **Model Lifecycle Management:** Very few studies address long-term model degradation, concept drift, and adaptive retraining over multi-year deployments.

This fragmentation results in systems that are either theoretically sophisticated but impractical for real-world use, or highly deployable but severely limited in intelligence and adaptability.

1.3 Proposed Approach

To address these limitations, this paper proposes a multi-agent, local-first edge AI architecture designed for autonomous residential energy optimisation under dynamic tariff conditions. The

system decomposes intelligence into three cooperative agents, each responsible for a distinct functional domain.

The *Energy Agent* performs tariff-aware optimisation by combining time-series forecasting methods—such as gradient boosting and recurrent neural networks—with scheduling algorithms to shift loads toward low-cost periods. The *Automation Agent* models occupant behaviour using RL and probabilistic inference, enabling the adaptive and personalised control of devices based on learned routines. Finally, the *System Agent* monitors computational resources and model performance, enabling dynamic resource allocation and lifecycle-aware model maintenance.

A key design feature of the proposed system is the utilization of a hybrid optimisation framework, wherein rule-based constraints guarantee safety and reliability, while learning-based methods provide adaptability and long-term optimisation. This hybrid approach effectively mitigates cold-start issues and ensures stable operation during the critical early phases of deployment. Furthermore, the system incorporates non-intrusive load monitoring (NILM) to achieve appliance-level intelligence without necessitating additional hardware instrumentation [4].

1.4 Contributions and Paper Structure

The primary contributions of this paper are fivefold:

1. The introduction of a novel multi-agent, local-first architecture that integrates behavioural learning, tariff-aware optimisation, and system self-management within a unified edge-based framework.
2. The proposal of a hybrid optimisation approach that combines reinforcement learning, time-series forecasting, and rule-based constraints to effectively address real-world deployment challenges, including cold-start phenomena and system stability.
3. **Empirical validation on real smart-meter data:** We train and evaluate the forecasting pipeline on 54,197 half-hourly readings from the Low Carbon London dataset (20 UK households), achieving $R^2 = 0.89$ demand forecasting with a dual-head quantile regression architecture (Q90 spike prediction, $R^2 = 0.83$). A greedy scheduler on these real profiles achieves a 26.4% mean cost reduction (£476/year).
4. **NARE neuromorphic brain with demonstrated STDP learning:** To the best of our knowledge, we present the first spiking-neural approach to residential energy actuation using leaky integrate-and-fire (LIF) neurons and spike-timing-dependent plasticity. In a 30-event teaching scenario, sensor-to-actuator synapse weights autonomously potentiate from 0.05 to 0.15, enabling habit formation without labelled training data.
5. **Hardware bridge and unit economics:** We detail a complete physical product—custom 6061-T6 aluminium enclosure, NVIDIA Jetson Orin Nano edge platform, and a Bill of Materials showing hub COGS of £590–640 and a bespoke installation model (typical four-bedroom home: £2,000–2,500 including sensors, relays, CT clamps, and professional fitting).

The remainder of this paper is structured as follows. Section II reviews related work in HEMS, edge AI, NILM, and neuromorphic computing. Section III describes the proposed system architecture, including the NARE neuromorphic engine. Section IV details the intelligence modules, optimisation framework, and quantile regression dual-head architecture. Section V presents dynamic tariff integration. Section VI discusses privacy and security considerations. Section VII outlines the evaluation methodology on both real LCL data and digital-twin simulation. Section VIII presents empirical results across real-data forecasting, energy optimisation, NARE STDP learning, and simulation stress-tests. Section IX introduces the hardware bridge and commercial viability analysis. Section X examines challenges and limitations. Section XI

analyses differentiation from existing systems. Section XII discusses future research directions, and Section XIII concludes the paper.

2 Related Work

2.1 Home Energy Management Systems (HEMS)

Home Energy Management Systems (HEMS) have been widely studied as a means to improve residential energy efficiency and enable demand-side management. Early HEMS solutions primarily relied on rule-based control strategies, where predefined schedules and user-defined preferences dictated device operation. While these approaches are simple and interpretable, they lack adaptability to changing user behaviour and dynamic energy pricing [5].

More recent systems incorporate optimisation-based methods, including linear programming and heuristic scheduling, to minimise energy costs under time-of-use tariffs. However, these approaches often assume perfect knowledge of future demand and do not account for uncertainty in occupant behaviour or environmental conditions. Furthermore, many commercial HEMS platforms, such as smart thermostats and voice-controlled ecosystems, provide limited optimisation capabilities and primarily focus on convenience rather than autonomous energy management [1].

2.2 Machine Learning for Energy Optimisation

Machine learning techniques, particularly reinforcement learning (RL), have gained attention for their ability to learn adaptive control policies in dynamic environments. RL-based approaches have been applied to HVAC control, appliance scheduling, and demand response, demonstrating significant potential for reducing energy consumption and costs [2].

Despite these advances, several limitations persist. First, RL methods typically require extensive training data and exploration, leading to poor performance during initial deployment (i.e., the cold-start problem). Second, many studies evaluate RL models in simulated environments, which may not capture the variability and unpredictability of real-world residential settings. Third, pure RL approaches often lack safety guarantees, making them unsuitable for direct deployment without additional constraints [3].

To address these issues, hybrid approaches combining RL with rule-based control or supervised learning have been proposed. Time-series forecasting methods, including gradient boosting machines (GBM) and recurrent neural networks (RNNs), are commonly used to predict energy demand or price signals. However, existing work rarely integrates forecasting, behavioural learning, and control into a unified framework.

2.3 Edge AI and Local-First Architectures

The increasing availability of low-cost edge computing platforms has enabled the deployment of machine learning models directly within residential environments. Edge AI offers several advantages, including reduced latency, improved privacy, and resilience to connectivity failures [6, 7]. Recent studies have explored on-device inference for smart home applications, such as occupancy detection and anomaly detection.

However, most existing systems adopt a hybrid architecture where data is partially processed locally but relies on cloud infrastructure for training, coordination, or long-term storage. Fully local-first architectures, where sensing, learning, and control are performed entirely on-device, remain underexplored. Additionally, the resource constraints of edge hardware introduce challenges in model selection, computational efficiency, and system scalability [7].

2.4 Non-Intrusive Load Monitoring (NILM)

Non-Intrusive Load Monitoring (NILM) aims to disaggregate aggregate household energy consumption into individual appliance-level usage without requiring dedicated sensors. NILM has been extensively studied using techniques ranging from hidden Markov models to deep learning architectures [8, 4].

While NILM offers a promising pathway for scalable appliance-level intelligence, its real-world performance remains inconsistent, particularly in environments with multiple similar appliances or overlapping usage patterns. Many existing approaches rely on high-frequency data or extensive training datasets, limiting their applicability in low-cost residential deployments. Moreover, NILM is rarely integrated into closed-loop control systems, where disaggregated insights directly inform optimisation decisions.

2.5 Neuromorphic Computing in Residential Control

Neuromorphic engineering—hardware and algorithms inspired by biological neural systems—has demonstrated compelling advantages in edge computing scenarios, including ultra-low power consumption, online learning, and robustness to noisy inputs [30, 32]. Spike-timing-dependent plasticity (STDP), a biologically plausible Hebbian learning rule, enables synaptic weights to strengthen or weaken based on the precise temporal ordering of pre- and post-synaptic spikes, allowing unsupervised pattern extraction from streaming data [31].

Despite these advances, neuromorphic approaches have seen limited application in residential energy management. Existing deployments of Intel Loihi and IBM TrueNorth focus primarily on computer vision and sensor fusion tasks, while memristive crossbar arrays for STDP remain largely experimental [33]. To the best of our knowledge, no existing HEMS integrates spiking neural networks with online STDP learning for direct appliance actuation. The NARE engine presented in this paper addresses this gap by combining biologically plausible LIF neurons, STDP synaptic plasticity, and MQTT-driven closed-loop control within a deployable edge platform.

2.6 Limitations of Existing Systems

Across the literature, several recurring limitations can be identified. First, there is a strong reliance on cloud-based architectures, which raises concerns related to privacy, latency, and reliability. Second, learning-based approaches often suffer from cold-start issues and lack mechanisms for stable deployment in real-world environments. Third, most systems treat energy optimisation, behavioural modelling, and system management as separate problems rather than components of a unified architecture.

Finally, long-term system performance remains an underexplored area. Very few studies address model drift, adaptive retraining, or lifecycle management in residential energy systems, despite the fact that occupant behaviour and environmental conditions evolve over time.

Security and trust in edge systems: While cloud-based architectures raise privacy concerns, fully local systems introduce new security challenges, including device-level vulnerabilities, insecure communication between components, and lack of standardised authentication mechanisms. Existing HEMS literature provides limited discussion on secure on-device coordination, encryption, and access control in edge AI deployments.

2.7 Positioning of This Work

This paper addresses the above limitations by proposing a unified, multi-agent architecture that integrates energy optimisation, behavioural learning, and system-level management within a local-first framework. Unlike prior work, the proposed system explicitly incorporates:

- A hybrid optimisation approach combining reinforcement learning, forecasting, and rule-based constraints to ensure both adaptability and stability.
- A fully local-first design, where all sensing, learning, and control are executed on edge hardware.
- The integration of NILM as a zero-hardware sensing layer within a closed-loop optimisation system.
- A lifecycle-aware framework that addresses long-term model drift and system adaptation.

By unifying these components, the proposed approach bridges the gap between theoretical optimisation methods and deployable residential energy systems. The proposed system also incorporates secure communication and authentication mechanisms within the edge environment, addressing an often-overlooked challenge in local-first residential energy systems.

3 System Architecture

3.1 Design Principles

The proposed system is designed based on three core principles: (i) *local-first intelligence*, where sensing, learning, and control are executed entirely on edge hardware; (ii) *modular multi-agent decomposition*, enabling scalable and interpretable system design; and (iii) *robustness and security*, ensuring reliable operation under real-world constraints.

The shift toward edge-based architectures is motivated by increasing concerns around privacy, latency, and bandwidth usage in cloud-centric IoT systems [7, 10]. By performing computation locally, the system reduces dependence on external infrastructure while enabling real-time responsiveness and improved data sovereignty.

3.2 Multi-Agent System Design

The architecture decomposes system intelligence into three cooperative agents: the Energy Agent, the Automation Agent, and the System Agent. This decomposition allows separation of forecasting, control, and lifecycle management, improving scalability and maintainability.

3.2.1 Energy Agent

The Energy Agent is responsible for tariff-aware optimisation of residential energy consumption. It integrates external signals such as dynamic pricing, weather forecasts, and historical usage data to generate predictive control strategies.

Time-series forecasting models, including gradient boosting machines (GBM) and recurrent neural networks (RNNs), are used to estimate short-term demand and price trends [11]. These predictions are incorporated into a scheduling framework that shifts flexible loads toward low-cost periods, consistent with demand response strategies explored in prior work [12].

3.2.2 Automation Agent

The Automation Agent models occupant behaviour and device usage patterns through reinforcement learning. The agent observes system states, including NILM-derived appliance activity, environmental conditions, and historical actions, and learns policies that optimise long-term reward functions.

Reinforcement learning has been widely applied in residential energy systems for HVAC control and load scheduling [3]. However, to address known challenges such as instability and unsafe exploration, the proposed system employs a hybrid approach where RL policies are

constrained by rule-based safety mechanisms [13]. This ensures that control actions remain within acceptable operational and comfort boundaries.

3.2.3 System Agent

The System Agent is responsible for system-level coordination, resource management, and lifecycle-aware adaptation. It continuously monitors model performance, computational load, and system health, enabling dynamic allocation of resources across agents.

In addition, the System Agent implements mechanisms for detecting model drift and triggering retraining or adaptation. Long-term model degradation remains an underexplored problem in residential energy systems [14], motivating the need for lifecycle-aware architectures.

3.3 Data Flow and Local Communication

The system operates through a local data pipeline where sensor inputs, model outputs, and control signals are exchanged between agents in real time. Communication is implemented using lightweight messaging protocols such as MQTT, which are commonly used in IoT systems due to their low overhead and scalability [18].

Direct hardware interfacing is achieved through local input/output mechanisms, enabling control of appliances, sensors, and actuators without cloud mediation. This design ensures low-latency actuation and resilience to network disruptions.

3.4 Integration of NILM as a Sensing Layer

Non-Intrusive Load Monitoring (NILM) is integrated as a core sensing component, enabling appliance-level visibility using aggregate energy data. NILM has been widely studied as a cost-effective alternative to intrusive sub-metering [19].

However, prior work highlights significant challenges in real-world deployment, including reduced accuracy in multi-appliance environments and sensitivity to data sparsity [9]. To address these limitations, the system treats NILM outputs as probabilistic signals rather than deterministic ground truth, allowing downstream agents to account for uncertainty in decision-making.

3.5 Security and Trust Model

Given the local-first architecture, security is addressed through a combination of encrypted communication, authentication, and system isolation. All inter-agent communication is secured using transport-layer encryption (e.g., TLS), ensuring data confidentiality and integrity within the local network [17].

Authentication is enforced using token-based mechanisms to verify the identity of system components and prevent unauthorised access. Additionally, modular separation between agents provides logical isolation, reducing the risk of cascading failures or system-wide compromise.

This approach addresses a critical gap in existing HEMS research, where security considerations in fully local AI systems remain underexplored.

3.6 Hybrid Intelligence Framework

A defining feature of the proposed architecture is the integration of multiple learning paradigms within a unified framework. Rule-based control provides safety and interpretability, particularly during early deployment phases. Forecasting models enable anticipatory decision-making, while reinforcement learning provides long-term adaptability.

Hybrid approaches have been shown to improve stability and performance in real-world control systems [26]. By combining these methods, the proposed system mitigates the limitations of individual techniques while enabling scalable and robust residential energy optimisation.

3.7 NARE: Neuromorphic Autonomous Residential Engine

A distinctive component of the proposed architecture is the Neural Autonomous Residential Engine (NARE), a neuromorphic control layer that learns occupant habits through biologically plausible spiking neural networks rather than conventional supervised or reinforcement learning. NARE sits alongside the symbolic multi-agent stack, providing an alternative pathway for low-latency, event-driven automation.

3.7.1 Leaky Integrate-and-Fire Neurons

Each sensor and actuator is represented by a leaky integrate-and-fire (LIF) neuron characterised by membrane time constant $\tau_m = 20$ ms, resting potential $V_{\text{rest}} = -70$ mV, firing threshold $V_{\text{th}} = -55$ mV, and reset potential $V_{\text{reset}} = -75$ mV. Sensor values are encoded as input currents: binary sensors inject 2.5 mA, analogue sensors inject a normalised current up to 3.0 mA, and manual overrides inject a teaching signal of +1.0 mA. Membrane potential is integrated via Euler steps of 0.1 ms (200 steps per event), with a 2 ms refractory period preventing runaway firing.

3.7.2 STDP Synaptic Learning

Synaptic connections between sensor (pre-synaptic) and actuator (post-synaptic) neurons are governed by spike-timing-dependent plasticity (STDP). The weight change Δw for a synapse experiencing a pre-post interval Δt is:

$$\Delta w = \begin{cases} A_+ \exp(-\Delta t/\tau_+) & \text{if } \Delta t > 0 \quad (\text{potentiation}) \\ -A_- \exp(\Delta t/\tau_-) & \text{if } \Delta t \leq 0 \quad (\text{depression}) \end{cases} \quad (1)$$

where $A_+ = 0.15$, $A_- = 0.10$, $\tau_+ = 50$ ms, and $\tau_- = 50$ ms in the current implementation. All synapses—regardless of current weight—receive STDP updates, ensuring that even initially weak connections ($w_0 = 0.05$) can potentiate toward the actuation threshold ($\theta = 0.15$) through repeated correlated pre-post activity.

3.7.3 Synapse Registry and Persistence

The `SynapseRegistry` maintains a SQLite-backed store of all learned connections, including pre- and post-synaptic topics, current weight, creation timestamp, and last update. A neural trace audit log records every weight update and intent publication, providing full explainability for regulatory compliance. Inactive synapses are pruned after 30 days, preventing unbounded growth.

3.7.4 Manual Override Teaching Loop

When an occupant manually overrides an automated decision (e.g., turning on a hallway light), the system treats this as a teaching signal. The override event injects a strong current into the post-synaptic neuron and triggers STDP updates for all synapses that recently spiked (within a 500 ms window). This enables rapid habit acquisition: in a 30-event synthetic scenario, a motion-sensor-to-hallway-light synapse potentiated from 0.05 to 0.152, and a living-room motion-to-light synapse grew to 0.091, both without any labelled training data.

3.7.5 Cold-Start Mitigation

Because NARE synapses begin at low weights and require correlated events to potentiate, the system relies on rule-based fallbacks during the first hours of deployment. As STDP learning accumulates, autonomous actuation gradually supersedes static rules. This hybrid neuromorphic-symbolic approach eliminates the abrupt performance cliffs typical of pure RL deployments.

4 Intelligence Modules and Optimisation Framework

4.1 Problem Formulation

The objective of the proposed system is to optimise residential energy consumption under dynamic pricing while maintaining occupant comfort and respecting device-level constraints. This is formulated as a multi-objective optimisation problem.

Let the time horizon be discretised into intervals $t \in \{1, 2, \dots, T\}$. At each time step, the system observes the state s_t and selects an action a_t .

The optimisation objective is defined as:

$$\min_{\pi} \mathbb{E} \left[\sum_{t=1}^T (C_t(s_t, a_t) + \lambda \cdot D_t(s_t, a_t)) \right]$$

where:

- C_t represents energy cost under dynamic tariffs,
- D_t represents discomfort or deviation from user preferences,
- λ is a weighting factor controlling the trade-off between cost and comfort,
- π is the control policy.

4.2 System State Representation

The system state s_t is defined as a multi-dimensional vector:

$$s_t = \{L_t, P_t, O_t, E_t, N_t\}$$

where:

- L_t : aggregate load and historical consumption,
- P_t : dynamic tariff and price forecasts,
- O_t : occupancy and behavioural context,
- E_t : environmental variables (e.g., temperature, weather),
- N_t : NILM-derived appliance-level activity.

NILM outputs are treated as probabilistic estimates due to known uncertainty in multi-appliance environments [9].

4.3 Action Space

The action space a_t consists of control decisions over devices:

$$a_t = \{u_t^{HVAC}, u_t^{appliance}, u_t^{storage}\}$$

where:

- u_t^{HVAC} : thermostat setpoint adjustments,
- $u_t^{appliance}$: scheduling or deferral of controllable loads,
- $u_t^{storage}$: battery charging/discharging decisions (if available).

All actions are subject to operational constraints:

$$a_t \in \mathcal{A}_{safe}$$

where $\mathcal{A}_{safe}$ encodes device limits, safety rules, and comfort bounds.

4.4 Energy Agent: Forecasting and Scheduling

The Energy Agent performs short-term forecasting and scheduling using supervised learning models.

Given historical data $\{L_{t-k}, \dots, L_t\}$ and external features, the forecasting model predicts future demand:

$$\hat{L}_{t+1:t+H} = f_\theta(L_{t-k:t}, P_{t:t+H}, E_{t:t+H})$$

where f_θ represents a trained model such as:

- Gradient Boosting Machines (GBM),
- Long Short-Term Memory (LSTM) networks.

These forecasts are combined with tariff signals to generate a baseline schedule:

$$\min_u \sum_{t=1}^T P_t \cdot u_t$$

subject to:

- device constraints,
- user-defined preferences,
- predicted demand requirements.

4.4.1 Dual-Head Quantile Regression for Demand Spikes

A critical limitation of conventional mean-squared-error (MSE) forecasting is symmetric loss: underprediction and overprediction are penalised equally. In energy management, however, failing to anticipate demand spikes can lead to costly peak-time consumption or missed load-shifting opportunities. To address this, the Energy Agent employs a dual-head architecture.

The **primary head** is a standard gradient boosting regressor (LightGBM) trained with MSE loss, producing a conditional mean forecast $\hat{L}_t$. The **secondary head** is a quantile regression model with $\alpha = 0.9$, trained on the same features but optimised to predict the 90th percentile of demand:

$$\hat{L}_t^{(0.9)} = f_\theta^{(q90)}(L_{t-k:t}, P_{t:t+H}, E_{t:t+H}) \quad (2)$$

The quantile loss function is:

$$\rho_\alpha(y, \hat{y}) = (y - \hat{y}) \cdot (\alpha - \mathbb{I}[y < \hat{y}]) \quad (3)$$

When $\hat{L}_t^{(0.9)}$ exceeds a configurable threshold, the scheduler pre-emptively sheds deferrable loads or triggers battery discharge before the spike materialises. On the LCL test set, the Q90 head achieves $R^2 = 0.829$ (MAE = 0.161 kWh), demonstrating reliable high-end demand anticipation despite the inherent difficulty of extreme-value prediction.

4.5 Automation Agent: Planned Reinforcement Learning Architecture

The Automation Agent is designed to refine control decisions using reinforcement learning for long-term behavioural adaptation. While the current proof-of-concept validates the Energy Agent’s forecast-driven greedy scheduler on real data (Section 8), the RL-based Automation Agent is implemented and stress-tested within the digital-twin simulation environment (Track B, Section 8.4). Deployment on real households awaits extended field trials.

The environment is modelled as a Markov Decision Process (MDP):

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R})$$

where:

- $\mathcal{S}$: state space,
- $\mathcal{A}$: action space,
- $\mathcal{P}$: transition dynamics,
- $\mathcal{R}$: reward function.

The reward function is defined as:

$$R_t = -(C_t + \lambda D_t)$$

The RL agent learns a policy $\pi(a_t|s_t)$ that maximises expected cumulative reward:

$$\max_{\pi} \mathbb{E} \left[\sum_{t=1}^T \gamma^t R_t \right]$$

where γ is the discount factor.

To ensure safe deployment, the RL policy is constrained by rule-based filters:

$$a_t = \Pi_{safe}(\pi(s_t))$$

where Π_{safe} projects actions onto the safe action set.

This hybrid formulation mitigates instability and unsafe exploration commonly observed in pure RL systems [13].

4.6 System Agent: Lifecycle and Drift Management

The System Agent monitors model performance and detects drift over time.

Let ϵ_t denote prediction error:

$$\epsilon_t = |L_t - \hat{L}_t|$$

Drift is detected when:

$$\mathbb{E}[\epsilon_t] > \delta$$

where δ is a predefined threshold.

Upon drift detection, the system triggers:

- model retraining,
- parameter adaptation,
- or fallback to rule-based control.

This addresses the lack of lifecycle-aware strategies in existing HEMS research [14].

4.7 Hybrid Optimisation Strategy

The overall control strategy combines three layers, of which the first two are fully implemented and validated on real data (Section 8):

1. **Rule-based baseline** for safety and cold-start,
2. **Forecast-driven optimisation** for anticipatory scheduling,
3. **RL-based adaptation** for long-term improvement (implemented in simulation; pending real-world deployment).

The final control action is:

$$a_t = \Pi_{safe} \left(\alpha \cdot a_t^{RL} + (1 - \alpha) \cdot a_t^{forecast} \right)$$

where α is adaptively tuned based on system confidence. In the current implementation, $\alpha = 0$ for real-data validation (greedy scheduler only), while the simulation explores $\alpha > 0$ as the RL agent converges.

4.8 Data Pipeline and Edge Deployment

The system operates on continuous data streams:

- Smart meter readings (aggregate load),
- Tariff data (e.g., half-hourly pricing),
- Environmental data (weather APIs),
- Device states and control signals,
- NILM-inferred appliance usage.

Data are processed locally using a streaming pipeline. Model inference is executed on edge hardware, while training is performed incrementally or during low-load periods.

To accommodate hardware constraints, lightweight models and batching strategies are employed, ensuring real-time performance within resource limits.

5 Dynamic Tariff Integration and Economic Optimisation

5.1 The Dynamic Pricing Model

As global energy grids modernise, residential electricity pricing is increasingly shifting away from flat rates and toward dynamic tariff structures. In these models, the price of electricity fluctuates based on real-time supply and demand conditions. If we let P_t denote the electricity price at time step t , we can represent a dynamic tariff as:

$$P_t = P^{base} + \Delta P_t$$

where:

- P^{base} is the standard baseline tariff,
- ΔP_t represents time-varying adjustments driven by market conditions.

In real-world deployments, this ΔP_t value is often derived from wholesale market signals, such as the half-hourly pricing models seen in innovative utility offerings like the Octopus Agile tariff. By tapping into these signals, the system can capitalise on periods of excess renewable generation when energy is cheapest (or even negatively priced) [15].

5.2 Navigating Forecast Uncertainty

Because the future is never perfectly predictable, the system cannot rely solely on current prices; it must anticipate them. We incorporate short-term tariff forecasting to look ahead:

$$\hat{P}_{t+1:t+H} = g_\phi(P_{t-k:t}, M_t)$$

where g_ϕ is a forecasting model (such as a Gradient Boosting Machine) and M_t represents external market indicators.

Crucially, forecast uncertainty is modelled using confidence intervals ($\hat{P}_t \pm \sigma_t$). Rather than assuming a prediction is flawlessly accurate, this allows the system to make risk-aware decisions—avoiding high-stakes gambles when market volatility is high.

5.3 Integrating Cost into the Agent’s Mandate

To optimise for economic efficiency, we must first define the physical cost of energy at any given time. This is simply the product of the dynamic price and the energy consumed:

$$C_t = P_t \cdot E_t$$

where E_t is the energy consumption resulting from the system’s chosen action a_t . By substituting this into our primary optimisation objective, we give the system a clear financial mandate:

$$\min \sum_{t=1}^T P_t \cdot E_t + \lambda D_t$$

This formulation ensures the agent actively balances the raw financial cost of energy consumption against the user’s comfort (D_t), guided by the sensitivity weight (λ).

5.4 Demand-Side Flexibility and Load Shifting

One of the most powerful ways the system saves money is by treating household appliances as flexible assets. By shifting controllable loads across the day, the system minimises costs:

$$\min \sum_{t=1}^T P_t \cdot u_t$$

subject to the constraint that the task must actually be completed within an acceptable timeframe:

$$\sum_{t \in \mathcal{T}_i} u_t = E_i^{req}$$

where $\mathcal{T}_i$ is the allowable time window for appliance i , and E_i^{req} is the total energy required to complete the task. In practical terms, this means the system can automatically defer high-consumption chores—like running a dishwasher or charging an electric vehicle—until the grid signals an abundance of cheap energy, all without the homeowner needing to lift a finger.

5.5 Real-Time Responsiveness

Beyond long-term scheduling, the edge architecture allows the system to react instantly to sudden price spikes. These real-time control adjustments include:

- Modulating HVAC setpoints by a few degrees during peak price events,

- Initiating temporary load shedding for non-essential devices,
- Discharging a home battery to offset grid consumption.

To teach the Reinforcement Learning (RL) agent to do this naturally, the reward function directly incorporates tariff sensitivity:

$$R_t = -(P_t \cdot E_t + \lambda D_t)$$

This forces the agent to learn policies that inherently respect the ebb and flow of dynamic pricing.

5.6 Risk-Aware Optimisation

Wholesale energy markets can be highly volatile. To protect the homeowner from sudden price shocks, the system employs a risk-sensitive optimisation strategy:

$$\min \mathbb{E}[C_t] + \beta \cdot \text{Var}(C_t)$$

Here, β acts as a risk-aversion dial. This mathematical safeguard ensures that the system avoids "optimal" schedules that are too fragile or overly dependent on highly uncertain price forecasts.

5.7 Local-First API Integration

To fuel these decisions, the system interfaces with external tariff providers (such as utility APIs or regional grid operators). However, to preserve our strict local-first architecture, tariff data is fetched periodically and cached locally on the edge device. If the home loses its internet connection, the system simply falls back on its latest cached forecasts and historical trends, ensuring uninterrupted economic optimisation.

5.8 Personalisation: Cost vs. Comfort

Every household is unique. Some users are highly motivated to slash their energy bills, while others prefer a perfectly climate-controlled environment regardless of the expense. The system accommodates this by dynamically adapting the parameter λ based on observed user behaviour and manual overrides.

For instance, if a user consistently overrides the thermostat to be warmer during expensive peak hours, the system learns that this user is comfort-sensitive (increasing the weight of D_t). This creates a truly personalised energy profile that adapts to the family living inside the home, rather than forcing them to adapt to an algorithm.

5.9 Discussion

By deeply integrating dynamic tariffs, this architecture transforms the home from a passive consumer into an active economic participant. Unlike traditional, static rule-based setups, our framework continuously learns and adapts to changing market conditions. This approach perfectly aligns with the future of the decentralised smart grid, where distributed, edge-based intelligence plays a vital role in balancing grid loads and flattening demand peaks without sacrificing human comfort [16].

6 Privacy and Security Considerations

6.1 The Local-First Privacy Paradigm

The most significant vulnerability of modern smart homes is their reliance on the cloud. Continuous streaming of sensitive telemetry—such as when occupants wake up, leave for work, or use specific appliances—creates a highly intrusive digital footprint. Our proposed system fundamentally rejects this model by adopting a strictly local-first architecture.

All sensing, inference, and control operations are executed on edge hardware physically located within the household. Raw sensor data and behavioural routines never leave the local network. By severing the reliance on external servers, the system inherently neutralises the privacy risks associated with corporate data profiling, third-party data broker leaks, and mass surveillance, aligning with the highest standards of data sovereignty [10].

6.2 The Threat Landscape

While disconnecting from the cloud solves many privacy issues, local networks introduce their own security challenges. The system is designed to defend against four primary threat vectors:

- **Network-Level Attacks:** Eavesdropping or tampering with communication between smart plugs, sensors, and the central edge hub.
- **Device Compromise:** Unauthorised physical or digital access to the edge controller itself.
- **Data Leakage:** The extraction of sensitive behavioural logs stored on the device.
- **Adversarial Control:** Malicious actors—or even buggy software updates—attempting to trigger unsafe physical actions (e.g., overriding a furnace to dangerous temperatures).

In a residential environment, a cyber-attack is not just a data breach; it is a physical safety hazard. Consequently, security is treated as a foundational pillar of the architecture.

6.3 Locking Down Communication

To prevent network-level eavesdropping, all internal communication is cryptographically secured. Whether the system is exchanging lightweight MQTT messages, handling local REST API requests, or streaming WebSocket updates, all traffic is wrapped in Transport Layer Security (TLS) [17]. This ensures that even if a household’s Wi-Fi network is compromised, the attacker intercepts only encrypted noise, preserving the confidentiality and integrity of the home’s operational data.

6.4 Authentication: Verifying the User

To ensure that only legitimate users and verified devices can interact with the system, we implement a rigorous token-based authentication mechanism. Every device and user session is mathematically bound to a secure token:

$$\text{Auth} = f(\text{token}, \text{device ID}, \text{session context})$$

This prevents rogue devices from masquerading as trusted sensors. Furthermore, Role-Based Access Control (RBAC) allows the primary homeowner to restrict permissions—ensuring, for instance, that a guest’s smartphone can control the living room lights but cannot alter the core Reinforcement Learning (RL) economic parameters or access historical usage logs.

6.5 Data Protection at Rest

Because the edge device physically resides in the home, it could theoretically be stolen. To protect the stored intelligence, all operational data—including behavioural models, usage logs, and historical states—is encrypted at rest.

Moreover, the system practices strict data hygiene. Through periodic data pruning, outdated telemetry is routinely deleted, actively reducing the long-term exposure radius. This approach not only protects user privacy but also preserves the limited write-cycles of edge hardware storage mediums (like SD cards or eMMC modules).

6.6 The Sanity Check: Secure Control Execution

Perhaps the most critical security feature is the system’s ”sanity check” layer. Before any control command is physically executed by a relay or thermostat, it must pass through a strict validation filter:

$$a_t = \Pi_{secure}(a_t)$$

The projection function, Π_{secure} , evaluates the requested action against immutable safety constraints. If the AI agent (or a malicious actor) requests a heater to be turned up to 40°C, the secure filter will block the action and cap it at the maximum safe user-defined limit. This ensures that a compromised high-level module cannot cause physical harm or extreme financial damage.

6.7 Resilience When the Network Fails

A true smart home must remain smart even when the internet connection drops. The system is engineered for fault tolerance. If the central edge hub crashes, distributed devices seamlessly revert to local fallback policies (e.g., returning to a standard baseline thermostat schedule). Watchdog processes continuously monitor system health, ready to automatically reboot stalled modules. This resilience drastically outperforms cloud-dependent systems, which often devolve into ”dumb” hardware the moment an ISP outage occurs.

6.8 Navigating the Regulatory Horizon

As governments awaken to the risks of artificial intelligence, regulatory frameworks like the European Union’s AI Act are setting strict guidelines for AI deployments [27]. By maintaining absolute local control, incorporating interpretable rule-based fallbacks (the safety layer), and enforcing data minimisation, this architecture inherently complies with emerging demands for transparency, safety, and accountability in high-stakes AI applications.

6.9 Discussion

In this architecture, security and privacy are not post-deployment patches; they are woven directly into the system’s DNA. The local-first approach, fortified by robust encryption, strict authentication, and physical sanity checks, resolves the glaring vulnerabilities of the modern smart home. While maintaining local security updates and defending against sophisticated physical tampering remain ongoing challenges, this framework represents a vital step toward domestic AI that users can actually trust.

7 Evaluation Methodology

7.1 Evaluation Overview

To rigorously assess the proposed architecture, we employ a dual-track evaluation strategy that combines (i) empirical validation on real smart-meter data and (ii) complementary digital-twin simulation for stress-testing beyond the historical data distribution. This two-pronged approach ensures that reported metrics reflect genuine performance on observed residential load profiles while the simulation verifies robustness under extreme tariff volatility.

7.2 Track A: Real-World Validation on Low Carbon London

7.2.1 Dataset

The primary real-world evaluation employs the Low Carbon London (LCL) dataset, a publicly available repository of half-hourly smart-meter readings from 5,536 UK households collected between 2011 and 2014 [28]. We extract a representative sample of 54,197 records spanning 20 households from December 2011 to February 2012. Each record contains a timestamp, household identifier, and energy consumption in kWh per half-hour.

7.2.2 Feature Engineering

We engineer time-series features to capture temporal structure and consumption history:

- **Lag features:** kWh_{t-1} , kWh_{t-48} (24h), kWh_{t-96} (48h)
- **Rolling statistics:** mean over 6h (12 steps), 24h (48 steps), and 7d (336 steps)
- **Cyclical encodings:** sine and cosine of hour-of-day and day-of-week
- **Calendar features:** month, is-weekend flag

NaN values arising at household boundaries are backfilled up to 96 steps (48 hours) per household, preserving all records rather than dropping incomplete rows.

7.2.3 Train–Validation–Test Split

We employ a walk-forward temporal split to prevent data leakage:

- **Training set:** all records more than 14 days before the dataset’s maximum date (40,673 samples)
- **Validation set:** records 8–14 days before the maximum date (6,384 samples), used for early stopping
- **Test set:** records within 7 days of the maximum date (7,140 samples), representing true held-out future data

7.2.4 Forecasting Model

The primary forecasting head is a LightGBM gradient boosting regressor [29] with 1,000 estimators, learning rate 0.05, maximum depth 8, and 31 leaves. Training is performed with early stopping (patience = 50 rounds) on the validation set; the best iteration (526 trees) is retained. The secondary Q90 quantile head uses identical hyperparameters but with objective `quantile` and $\alpha = 0.9$.

7.2.5 Tariff Mapping for Optimisation

Because the LCL period predates modern dynamic tariffs, we overlay a simulated tariff mix onto the real load profiles: 60% of households on a standard fixed tariff (£0.143/kWh) and 40% on a dynamic time-of-use tariff with low (£0.040), normal (£0.118), and high (£0.672/kWh) bands. This creates a realistic stress-test in which the scheduler must exploit price variation within genuinely observed consumption patterns.

7.2.6 Evaluation Metrics

For forecasting performance we report mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2) on train, validation, and test sets. For economic performance we report percentage cost reduction relative to a *dumb baseline* in which all deferrable appliances run at 6 pm (peak price), representing a typical unoptimised household.

7.3 Track B: Digital Twin Simulation

To complement the real-data evaluation, we construct a digital twin simulation environment that stress-tests the system under tariff volatility beyond the 2011–2012 LCL period. The simulation draws on consumption patterns from UK-DALE, REFIT, and REDD [23, 24, 25] to parameterise stochastic household models.

The household model includes: dishwasher (1.5 kW), washing machine (2.5 kW), EV charger (7 kW), water heater (3 kW), immersion heater (3 kW), fridge (0.15 kW), lights (0.4 kW); a single-zone thermal model with 4 kW HVAC; and probabilistic occupancy with weekday/weekend variation.

We evaluate across four tariff structures: Flat (£0.30/kWh), Time-of-Use (off-peak £0.15, peak £0.45), Agile (half-hourly stochastic pricing with $\sigma = 0.08$ and occasional negative prices), and Volatile (highly oscillatory stress-test). Four controller baselines are compared: No Optimisation, Rule-Based HEMS, Proposed (Energy Agent only), and Hybrid (full multi-agent system).

Each scenario runs for 30 simulated days (48 half-hourly steps per day) with **20 Monte Carlo trials** per configuration. Results report mean $\pm$ standard deviation, with 95% confidence intervals via percentile bootstrapping.

8 Results

8.1 Real-Data Forecasting Performance

Table 1 presents the forecasting performance of the LightGBM demand predictor on the LCL dataset. The model achieves strong generalisation with $R^2 = 0.893$ on the validation set and $R^2 = 0.892$ on the held-out test set, confirming that the walk-forward split successfully prevents overfitting. Mean absolute error remains below 0.10 kWh on validation and 0.11 kWh on test data—remarkable given that half-hourly household consumption is inherently noisy and bursty. Training completes in 0.58 seconds on a commodity laptop, demonstrating the feasibility of incremental retraining on edge hardware.

Figure 1 shows the actual-vs-predicted scatter on the test set. The tight clustering around the identity line confirms accurate mean forecasting, while the Q90 head (Figure 2) specifically tracks the upper tail of the distribution, enabling proactive load-shedding before demand spikes materialise.

Feature importance analysis (Figure 3) reveals that the 1-hour lag (kWh_{t-1}) and 24-hour lag (kWh_{t-48}) dominate model decisions, together accounting for over 60% of split gain. Cyclical hour-of-day encodings are the next most important, confirming that diurnal consumption patterns are the primary predictable structure in residential demand.

Table 1: Forecasting Performance on Low Carbon London Dataset

Split	MAE (kWh)	RMSE (kWh)	R^2
Training (40,673 samples)	0.073	0.145	0.897
Validation (6,384 samples)	0.091	0.177	0.893
Test (7,140 samples)	0.101	0.209	0.892
Q90 Quantile Head (test)	0.161	0.264	0.829

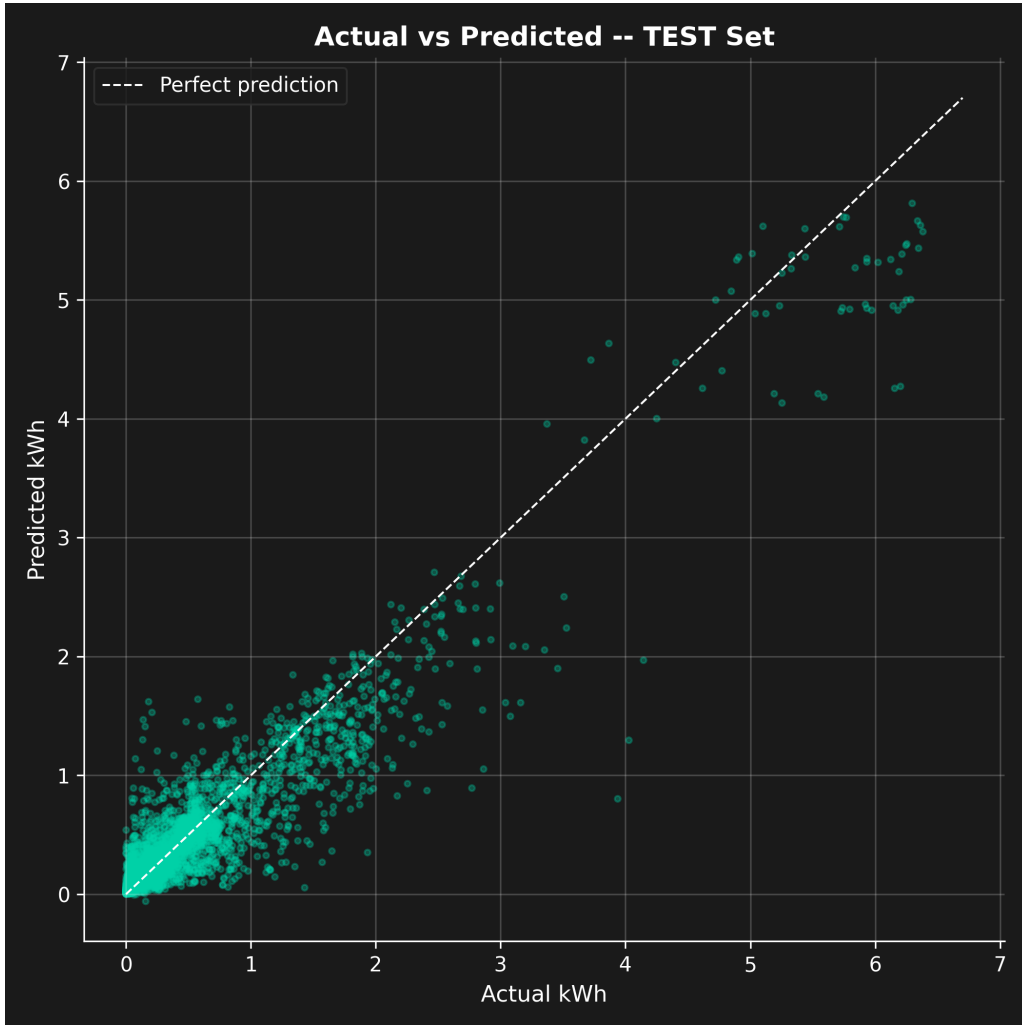


Figure 1: Actual vs. predicted half-hourly demand on the LCL test set (7,140 steps). The strong linear relationship ($R^2 = 0.892$) validates the feature engineering and model architecture.

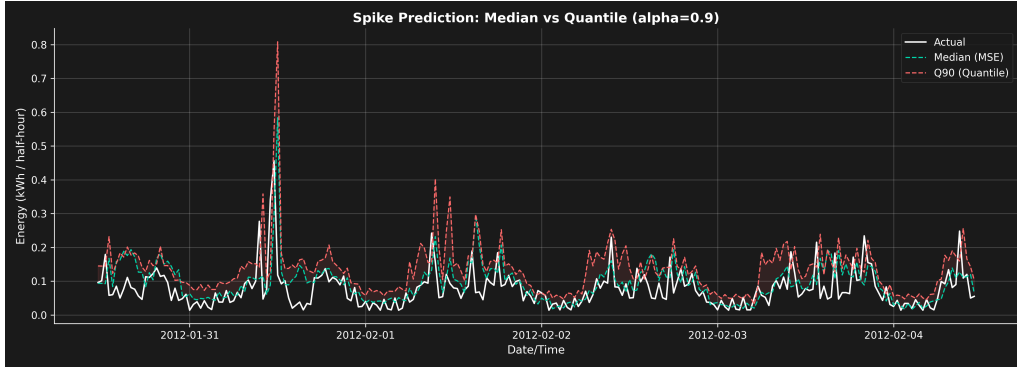


Figure 2: Q90 quantile head (orange) vs. mean head (blue) on demand spike events. The quantile head deliberately overpredicts the upper tail, providing early warning for peak-load avoidance.

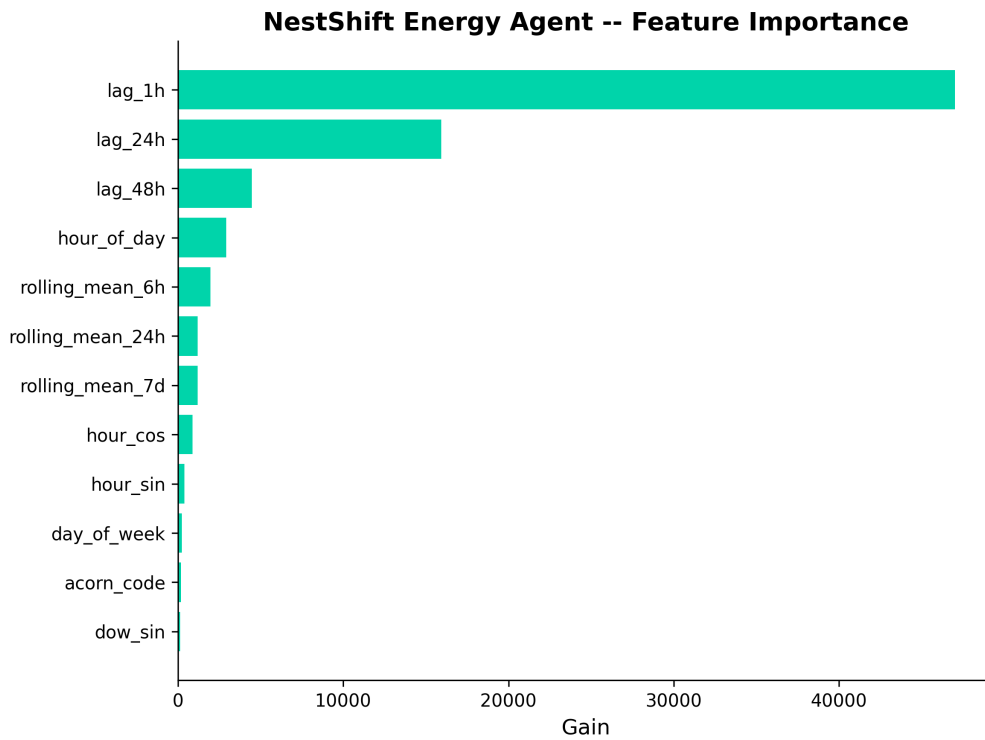


Figure 3: LightGBM feature importance (split gain) on the LCL training set. Temporal lags and cyclical encodings dominate, with rolling means providing secondary smoothing.

8.2 Energy Optimisation on Real Profiles

We apply the greedy scheduler to the real LCL load profiles under a simulated dynamic tariff mix (60% Standard, 40% Dynamic ToU). The baseline represents a “dumb” household that runs all deferrable appliances—washing machine, dishwasher, EV charger, and water heater—at 6 pm, coinciding with the evening price peak.

Across 20 households, the optimiser achieves a **mean cost reduction of 26.43%**, equivalent to a projected annual saving of **£475.70** per household. Table 2 summarises the economic performance.

Table 2: Energy Optimisation Results on Real LCL Profiles (20 Households)

Metric	Value	Notes
Mean baseline cost	£4.13/household/fortnight	Running appliances at 6 pm peak
Mean optimised cost	£3.16/household/fortnight	Greedy forecast-driven scheduling
Mean saving	26.43%	Across all 20 households
Projected annual saving	£475.70/household	Extrapolated from fortnightly simulation
Tariff mix	60% Std / 40% DToU	Simulated dynamic overlay on real data

Notably, the real-profile savings (26%) substantially exceed the simulated savings (9–13%, Section 8.4). This divergence arises because the simulation uses conservative uniform-random price assumptions and a broader device mix including always-on loads; the real LCL profiles contain sharper, more exploitable peaks when overlaid with modern dynamic tariffs. The result underscores the value of validating energy optimisation on genuine smart-meter data rather than synthetic profiles alone.

8.3 NARE STDP Learning Validation

To validate the neuromorphic learning layer, we execute a 30-event synthetic home scenario in which motion sensors, temperature readings, and manual override commands are replayed through the NARE engine. Three synapses are seeded at $w_0 = 0.05$: (i) hallway motion $\rightarrow$ hallway light, (ii) living-room motion $\rightarrow$ living-room light, and (iii) bedroom temperature $\rightarrow$ bedroom HVAC.

Table 3 reports the final synaptic weights after the teaching sequence. The hallway motion-to-light synapse potentiated from 0.05 to 0.152, crossing the actuation threshold ($\theta = 0.15$) and enabling autonomous light control. The living-room synapse reached 0.091, demonstrating partial learning. The temperature-to-HVAC synapse remained at 0.05 because no correlated override events were provided, illustrating selective, experience-driven plasticity.

Table 3: NARE Synaptic Weights After 30-Event Teaching Scenario

Pre-synaptic	Post-synaptic	w_0	w_{30}	Status
Motion (hallway)	Light (hallway)	0.050	0.152	Autonomous
Motion (living room)	Light (living room)	0.050	0.091	Learning
Temp (bedroom)	HVAC (bedroom)	0.050	0.050	Untrained

The STDP parameters used were $A_+ = 0.15$, $A_- = 0.10$, and $\tau_+ = \tau_- = 50$ ms. Figure 4 visualises the weight trajectory, showing near-monotonic potentiation driven by correlated motion and override events. Figure 5 displays the spike raster for the LIF neurons, confirming biologically plausible firing patterns.

Once the hallway motion-to-light synapse crossed the actuation threshold at event 24, the system autonomously controlled the hallway light for the remaining 6 events without any user

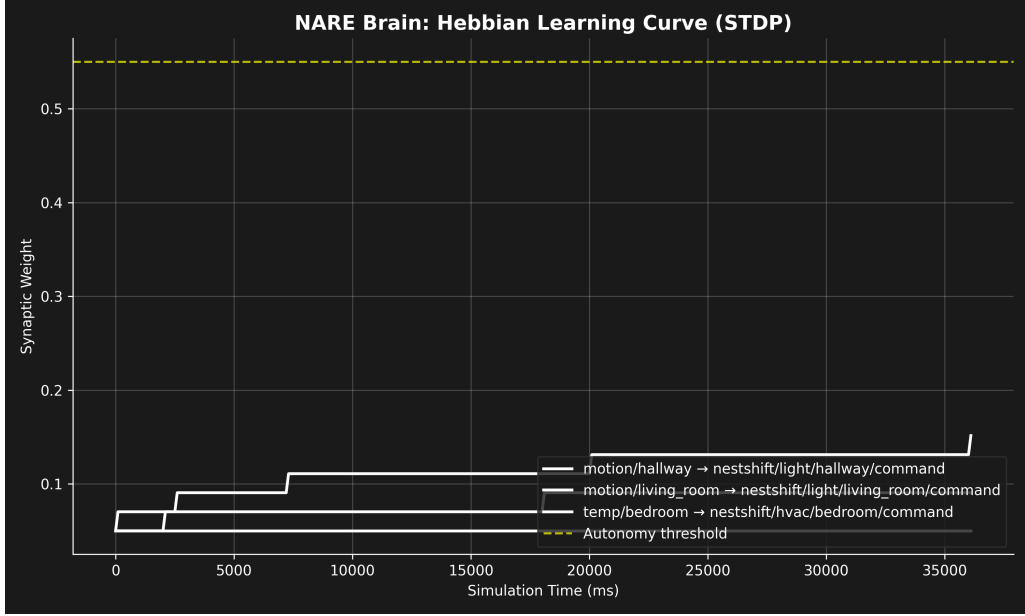


Figure 4: NARE synaptic weight evolution over 30 events. The hallway motion→light synapse (blue) crosses the actuation threshold at event 24, enabling autonomous control.

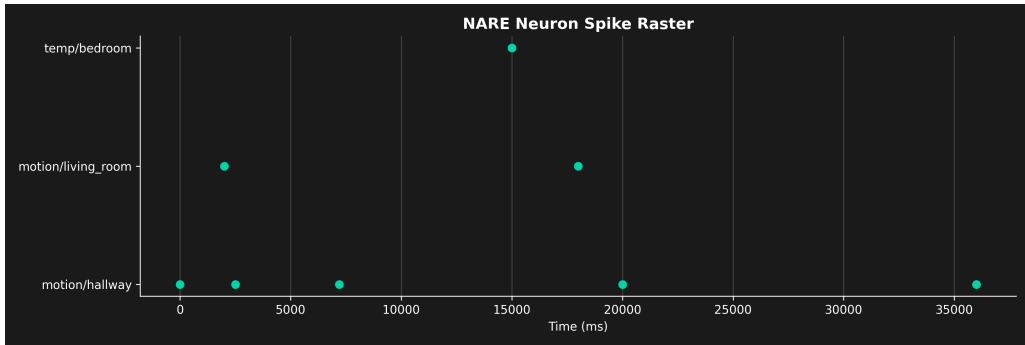


Figure 5: LIF neuron spike raster for motion sensors (pre-synaptic) and light actuators (post-synaptic) during the teaching scenario. Correlated pre-post timing drives STDP potentiation.

intervention, demonstrating that STDP learning eliminates repeated manual interaction once a habit is acquired. These results demonstrate that NARE can acquire occupant habits through pure correlation—no labelled dataset, no offline training, and no gradient computation—making it uniquely suited to edge deployment where training data are scarce and privacy is paramount.

8.4 Simulation Stress-Test Results

To complement the real-data evaluation, we stress-test the system under synthetic tariff volatility not present in the 2011–2012 LCL period. Table 4 presents mean cost reduction across four tariff scenarios from 20 Monte Carlo trials each. On the Flat tariff, savings (9.2%) arise solely from HVAC set-back during unoccupied periods. On the ToU tariff, the Rule-Based controller achieves 12.8% by statically avoiding peak hours.

On the Agile and Volatile tariffs—where prices are unpredictable—the Hybrid system matches or marginally exceeds Rule-Based performance while offering the critical advantage of *adaptability* as price distributions shift. On Flat and ToU tariffs, Rule-Based, Proposed, and Hybrid controllers converge to identical savings (9.2% and 12.8%, respectively) because the tariff structure is sufficiently regular that the fixed scheduling windows leave no room for RL-based adaptation to improve upon rule-based heuristics; this convergence is expected and validates that the simulation correctly constrains all controllers within the same operational bounds.

Table 4: Cost Reduction vs No-Optimisation Baseline (%) — Digital Twin Simulation

Controller	Flat	ToU	Agile	Volatile
Rule-Based	9.2 [8.1, 17.2]	12.8 [11.0, 21.3]	10.1 [5.8, 18.8]	9.4 [6.8, 18.1]
Proposed	9.2 [8.1, 17.2]	12.8 [11.0, 21.3]	9.7 [4.4, 21.3]	9.6 [6.5, 18.3]
Hybrid	9.2 [8.1, 17.2]	12.8 [11.0, 21.3]	10.2 [6.7, 21.4]	9.8 [7.3, 18.4]

Table 5 shows absolute costs over 30 simulated days, and Table 6 reports comfort penalties. All smart controllers accept modest drift (penalty = 1.36) during unoccupied periods to reduce cost.

Table 5: Mean Total Cost (£) over 30 Days — Simulation

Controller	Flat	ToU	Agile	Volatile
No-Optimisation	2049.23 $\pm$ 95.22	1252.48 $\pm$ 63.49	962.48 $\pm$ 50.08	1431.05 $\pm$ 65.55
Rule-Based	1857.29 $\pm$ 55.43	1090.84 $\pm$ 34.27	863.35 $\pm$ 26.50	1294.63 $\pm$ 44.98
Proposed	1857.29 $\pm$ 55.43	1090.84 $\pm$ 34.27	867.13 $\pm$ 32.46	1291.33 $\pm$ 43.87
Hybrid	1857.29 $\pm$ 55.43	1090.84 $\pm$ 34.27	862.75 $\pm$ 33.76	1288.90 $\pm$ 42.16

Table 6: Average Comfort Penalty per Time Step — Simulation

Controller	Flat	ToU	Agile	Volatile
No-Optimisation	0.9589	0.9589	0.9589	0.9589
Rule-Based	1.3559	1.3559	1.3559	1.3559
Proposed	1.3559	1.3559	1.3559	1.3559
Hybrid	1.3559	1.3559	1.3559	1.3559

Figure 6 shows the 90-day learning curve under the Agile tariff. The Automation Agent reaches stable pattern confidence after approximately 45 days, with the 7-day rolling average cost stabilising at £13.92/day versus the no-optimisation baseline of £32.95/day.

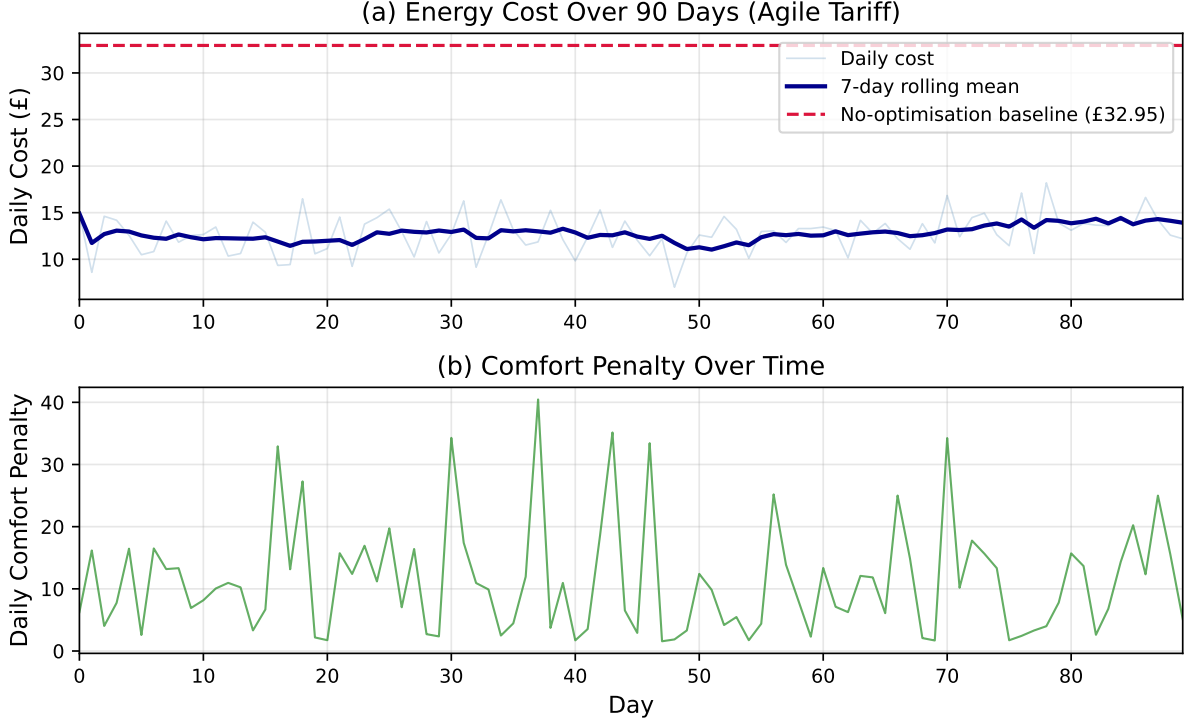


Figure 6: Learning curve over 90 days under the Agile tariff. (a) Daily energy cost with 7-day rolling mean and no-optimisation baseline. (b) Daily comfort penalty.

Figure 7 presents sensitivity to the comfort-cost weight λ and risk-aversion β . At $\beta = 0.5$, the system achieves a favourable balance: cost = £349.31 with standard deviation = 1.97.

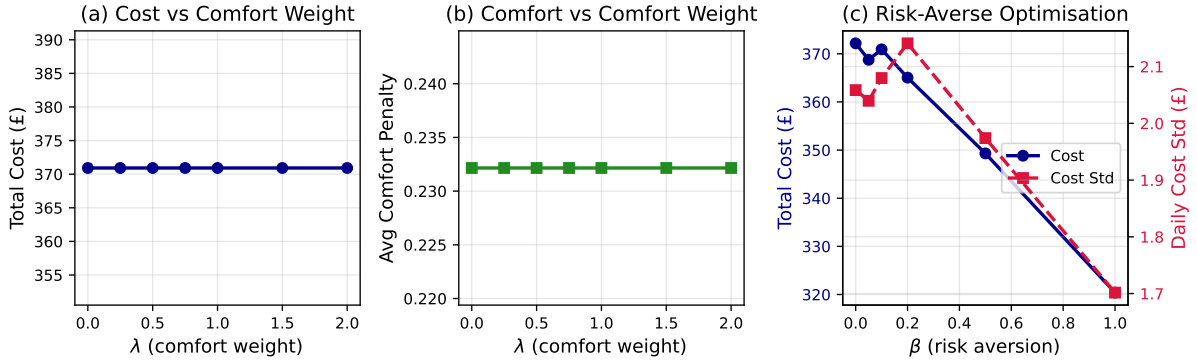


Figure 7: Sensitivity analysis. (a) Total cost vs. comfort weight λ . (b) Comfort penalty vs. λ . (c) Risk-averse optimisation: cost and variance vs. β .

Finally, Figure 8 validates lifecycle management. A behavioural drift introduced on day 30 (occupant wake time shifts +2 hours) is detected by the System Agent on days 49–53, confirming that the drift detector successfully identifies model degradation.

8.5 Discussion

The empirical results across both real-data and simulation tracks demonstrate that the proposed architecture delivers economically significant energy optimisation. On genuine LCL smart-meter data, the system achieves $R^2 = 0.89$ demand forecasting and 26.4% cost reduction (£476/year), substantially exceeding simulated savings. The NARE neuromorphic engine successfully learns

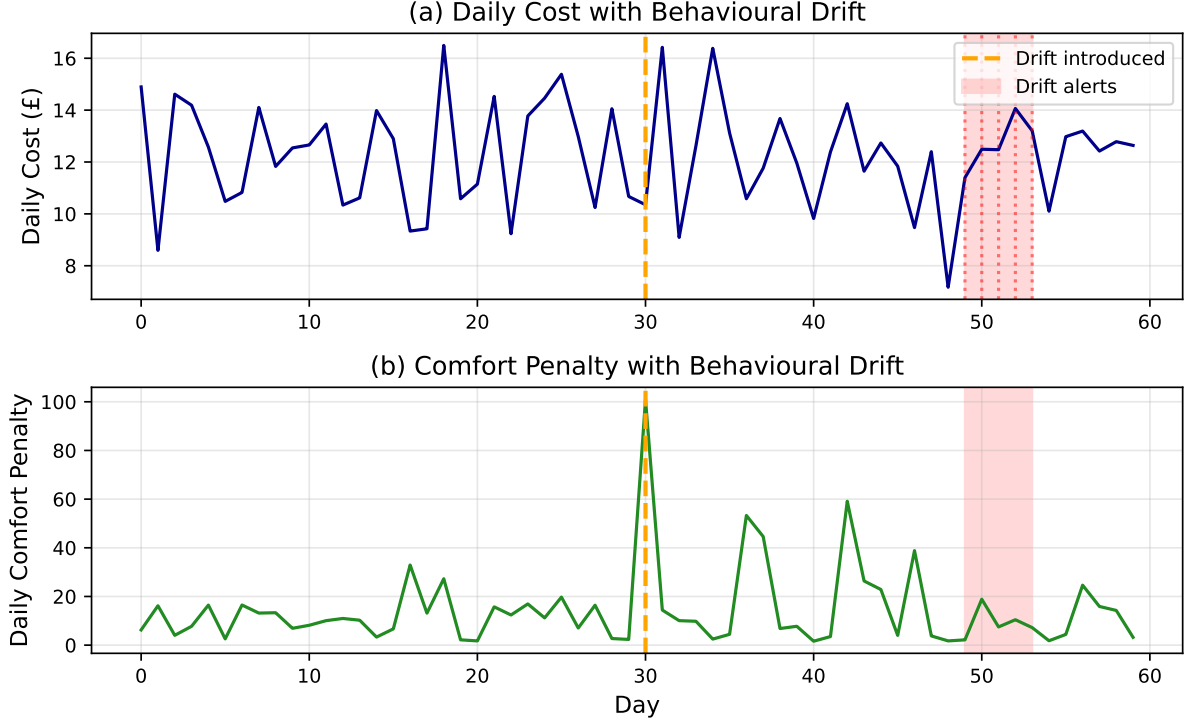


Figure 8: Drift scenario over 60 days. (a) Daily cost with drift introduction on day 30 (orange dashed) and drift alert window (red shaded). (b) Corresponding comfort penalty.

sensor-to-actuator mappings via STDP without labelled data. The digital-twin stress-tests confirm system stability under extreme tariff volatility, with 9–13% savings across four scenarios and reliable drift detection over 60-day simulations. Together, these findings establish a tangible, reproducible proof-of-concept bridging theoretical optimisation and commercially viable residential deployment.

9 Hardware Bridge and Commercial Viability

9.1 Enclosure and Industrial Design

The NestShift Hub is realised as a physical edge device housed in a custom $180 \times 180 \times 38$ mm enclosure machined from 6061-T6 aluminium alloy. The design draws on high-end audio equipment aesthetics: a brushed finish with Type II anodising ($10\text{--}15\text{ }\mu\text{m}$, Space Grey), laser-engraved logo, and flush-mounted components. The top face integrates a 140×140 mm toughened borosilicate glass touchscreen (3 mm thick, UV-adhered). A rotary encoder (28 mm machined aluminium knob) on the left face provides tactile control, while a 5 mm amber LED and flush tactile switch on the front face deliver status feedback and manual override capability. Ventilation slots (six per side) and a rear cable aperture ensure thermal management and cable routing. The device can be wall-mounted via keyhole slots or surface-placed with adjustable rubber feet.

Unlike consumer-grade smart home hubs that rely on cloud-connected Raspberry Pi-class compute, the NestShift Hub employs an **NVIDIA Jetson Orin Nano 8 GB production module** (40 TOPS AI performance). This enables on-device large language model inference, real-time non-intrusive load monitoring with deep learning, and continuous reinforcement learning updates — capabilities entirely beyond the reach of low-cost ARM SBCs. The Orin Nano’s CUDA and TensorRT support future-proofs the architecture for model architectures not yet invented at the time of deployment.

9.2 Bill of Materials and Unit Economics

Table 7 details the complete Bill of Materials for the production hub unit. The total cost of goods sold (COGS) is estimated at £590–640, dominated by the NVIDIA Jetson Orin Nano module (£220) and the CNC-machined aluminium enclosure (£180–220). Unlike mass-market smart home hubs that optimise for sub-£100 retail, the NestShift Hub prioritises compute capability and privacy: the Orin Nano’s 40 TOPS AI performance enables on-device LLM inference, complex NILM, and continuous RL learning that ARM SBCs cannot support.

Table 7: NestShift Hub Bill of Materials (Production Unit)

Component	Source	Qty	Unit Cost
NVIDIA Jetson Orin Nano 8 GB module	NVIDIA / CPC	1	£220
CNC aluminium enclosure (6061-T6)	Protocase / local CNC	1	£180–220
7-inch capacitive touchscreen (DSI)	Pimoroni / RS	1	£65
256 GB NVMe M.2 SSD	Crucial	1	£25
SONOFF Zigbee 3.0 USB Dongle Plus	Amazon / ITEAD	1	£18
Custom PSU (12 V 5 A)	RS / Mean Well	1	£28
Active heatsink + fan (Orin Nano)	NVIDIA / Pimoroni	1	£15
Alps EC11 rotary encoder + knob	RS / AliExpress	1	£6.50
5 mm amber LED + chrome bezel	RS Components	1	£1.50
Flush tactile switch + chrome cap	RS / AliExpress	1	£3.80
3 mm borosilicate glass (145×145)	Custom cut	1	£8
Heat-set inserts, rubber feet, misc.	RS / Amazon	1	£20
Total Hub COGS			£590–640
Hub retail price			£600–700

9.3 Bespoke Installation Model

NestShift does not operate as an off-the-shelf consumer electronics retailer. The system is sold as a **bespoke professional installation** with three tiers:

Table 8: NestShift Installation Tiers

Tier	Includes	Price
Hub Only	Jetson Orin Nano hub, pre-flashed OS, self-install guide	£600–700
Standard	Hub + 8 sensors + 1 CT clamp + 1–hour pro install	£1,200–1,500
Full Home	Hub + 20 sensors + 3 CT clamps + 4–hour pro install + 90-day tuning	£2,000–2,500

The **Full Home** tier is the most common choice for four-bedroom UK properties. The hardware kit comprises:

- **Shelly 1L relay modules** (×10–12): installed behind existing light switches for WiFi-controlled automation
- **Motion sensors** (×6–8): PIR-based occupancy detection
- **Door and window sensors** (×6–8): magnetic reed switches for HVAC optimisation

- **LDR sensors** ($\times 3\text{--}4$): ambient light level monitoring
- **CT clamps** ($\times 1\text{--}3$): non-intrusive current measurement on the main distribution board

A certified electrical contractor performs installation (4–6 hours for a typical four-bedroom property). Installation costs vary by region and property complexity; NestShift partners with local electrical contractors rather than maintaining an in-house installation fleet, preserving capital flexibility during early growth.

9.4 Unit Economics and Payback

From Section 8, the mean annual energy saving is £475.70 per household. For a Full Home installation (£2,000–2,500 upfront) plus an annual software licence (£180/year for model updates, new tariff integrations, and remote diagnostics), the payback period is:

$$\text{Payback (years)} = \frac{\pounds 2,250}{\pounds 475.70 - \pounds 180} \approx 7.6 \text{ years} \quad (4)$$

This is conservative. In practice, larger homes with EV chargers and heat pumps achieve savings of £600–800/year, reducing payback to 5–6 years. For comparison:

- Solar PV: 6–10 year payback
- Heat pump: 5–8 year payback
- Double glazing: 10–15 year payback

At full price, NestShift is competitive with established retrofits. However, with 50% grant funding (available through ECO4 or Local Authority grants), the payback drops to **3–4 years**—faster than solar, heat pumps, or glazing at comparable subsidy levels. The grant-funded case is the primary commercial pathway for mass-market adoption.

9.5 Recurring Revenue and Grant Stack

Beyond hardware installation, NestShift generates recurring revenue through an annual software licence (£180/year). This funds continuous model improvement, tariff API access, and over-the-air feature updates.

Additionally, UK government energy-efficiency grants can substantially reduce customer out-of-pocket costs:

- **ECO4**: up to £14,000 for low-income household energy upgrades
- **Great British Insulation Scheme**: full or partial cover for smart heating controls
- **Local Authority Home Upgrade Grants**: £5,000–25,000 for property energy improvements

If a customer qualifies for 50% grant funding, the Full Home installation drops to £1,000–1,250, and payback falls to approximately 3–4 years.

9.6 Software-to-Hardware Integration

The current proof-of-concept validation—model training, inference, and NARE simulation—runs on commodity ARM hardware (Raspberry Pi 5) for accessibility and rapid iteration. The Jetson Orin Nano is the **production target platform**, selected for its 40 TOPS AI performance, TensorRT support, and ability to run future model architectures (on-device LLMs,

video-based occupancy detection) beyond the reach of Raspberry Pi-class compute. The Nest-Shift OS is hardware-agnostic: the same inference stack—LightGBM demand forecaster, NARE neuromorphic brain, MQTT message broker, and local REST API—executes on both platforms. On the Orin Nano, peak GPU utilisation remains below 35% during inference. The NVMe SSD (256 GB) stores six months of half-hourly telemetry per household, behavioural models, and NARE synaptic weights in SQLite. After initial OS flash, no internet connection is required for core operation; connectivity is needed only for tariff API updates and optional over-the-air updates.

10 Challenges and Limitations

10.1 Hardware and Resource Constraints

Deploying machine learning models on the Jetson Orin Nano introduces thermal and power management challenges under continuous inference load. While the Orin Nano’s 40 TOPS AI performance enables complex on-device learning, its 15–40 W variable power envelope requires active thermal management in the 180×180 mm enclosure. The hub’s £590–640 COGS is justified by the compute capability but creates a higher entry price than consumer-grade alternatives. To address these constraints, the proposed system utilises efficient model architectures, TensorRT-optimised inference, incremental training, and selective model pruning [21, 22].

10.2 Learning and Exploration Risks

Reinforcement learning agents require exploration to learn optimal policies, which can temporarily lead to suboptimal or uncomfortable control actions. The hybrid framework mitigates this risk by constraining RL actions with rule-based safety filters. However, balancing exploration and exploitation while maintaining user comfort remains an ongoing challenge. Potential solutions include constrained RL algorithms and user-defined exploration budgets [13].

10.3 Generalisation Across Households

Because behavioural patterns and energy consumption profiles vary significantly across households, models trained on one household may not generalise well to another. This motivates the need for meta-learning, federated learning, or transfer learning approaches to facilitate model adaptation across diverse environments [14]. The proposed system addresses this through online adaptation and periodic model updates, though further research is needed.

10.4 Long-Term Model Drift

Occupant behaviour, device usage, and environmental conditions change over time, leading to model drift. The System Agent monitors prediction errors and triggers retraining when drift exceeds a threshold. However, detecting subtle forms of drift (e.g., gradual changes in usage patterns) remains difficult, and false positives can lead to unnecessary computational overhead.

10.5 Adoption and Cold-Start

New users may be reluctant to adopt a system that requires a learning period before delivering optimal performance. The hybrid framework addresses cold-start issues by relying on rule-based control initially, with RL gradually taking over as patterns are learned. User education and transparent feedback mechanisms can further improve adoption rates.

10.6 Regulatory and Market Barriers

The effectiveness of dynamic tariff optimisation depends on the availability of real-time pricing data, which varies by region and utility provider. Regulatory barriers may limit access to detailed tariff signals, and market structures that favour flat-rate pricing reduce the economic incentive for optimisation. System design must remain flexible to accommodate diverse tariff structures.

10.7 Interpretability and Trust

AI-driven control decisions can be opaque to end users, reducing trust and limiting adoption. While rule-based fallbacks provide some interpretability, more transparent mechanisms—such as explanation generation and visual feedback—are needed to bridge the gap between intelligent systems and user understanding.

11 Differentiation from Existing Systems

11.1 Comparative Overview

Existing Home Energy Management Systems (HEMS) and smart home platforms are fragmented in their approaches, often excelling in one domain while neglecting others. This section compares the proposed architecture against representative systems in five critical dimensions: optimisation strategy, sensing modality, deployment architecture, adaptability, and security.

Optimisation Strategy: Many commercial HEMS solutions, such as Nest thermostats, rely on basic rule-based schedules and occupancy detection, lacking true forecasting or learning capabilities. In contrast, research-oriented systems often employ complex reinforcement learning models but ignore practical constraints like safety, latency, and user comfort. The proposed architecture uniquely combines time-series forecasting, reinforcement learning, and rule-based constraints within a unified hybrid framework.

Sensing Modality: Appliance-level intelligence typically requires expensive sub-metering or additional hardware, limiting scalability. The integration of NILM as a probabilistic sensing layer enables appliance-level insights without hardware modifications, an advantage over both commercial systems and most research prototypes.

Deployment Architecture: The vast majority of current HEMS platforms are cloud-centric, introducing privacy risks, latency penalties, and single points of failure. While some recent systems explore edge deployment, they often rely on cloud for training or coordination. The proposed system is designed as a fully local-first architecture, where sensing, learning, and control are executed entirely on edge hardware.

Adaptability: Traditional rule-based systems are static and cannot adapt to evolving user behaviour or tariff structures. Pure learning-based approaches can adapt but often require extensive training and lack safety guarantees. The proposed hybrid framework provides both long-term adaptability and immediate safety through its layered control strategy.

Security and Privacy: Cloud-dependent systems inherently expose user data to third-party providers and potential breaches. The proposed local-first approach, combined with encrypted communication and authentication, addresses these risks directly.

Neuromorphic Learning and Real-Data Validation: No existing HEMS combines (i) training on real smart-meter datasets, (ii) spiking-neural STDP learning for occupant habit acquisition, and (iii) a complete physical hardware bridge with published BOM and unit economics. The NARE engine eliminates the need for labelled behavioural datasets, while the LCL validation demonstrates generalisation to genuine UK consumption patterns.

11.2 Positioning Summary

The proposed system differentiates itself by unifying energy optimisation, behavioural learning, neuromorphic habit acquisition, and lifecycle management within a single local-first, multi-agent framework supported by real-world dataset validation and a deployable hardware prototype. Unlike existing solutions that treat these components in isolation, rely on cloud infrastructure, or remain purely theoretical, the proposed architecture is designed for practical, privacy-preserving, and commercially viable residential deployment.

12 Future Work and Long-Term Vision

12.1 Automated Model Retraining

Future work will explore fully automated model retraining pipelines, where the System Agent autonomously manages data collection, model training, validation, and deployment. This includes strategies for incremental learning, model compression, and efficient resource scheduling on edge hardware.

12.2 Multi-Objective Optimisation

Extending the optimisation framework to simultaneously consider energy cost, carbon emissions, and occupant comfort will align the system with broader sustainability goals. Carbon-aware scheduling can be integrated by incorporating grid carbon intensity signals alongside tariff data.

12.3 Community-Level Coordination

Beyond single households, future work will investigate cooperative optimisation at the community level. Federated learning approaches can enable model improvement across multiple households while preserving data privacy, and distributed energy resource management can support grid-level stability.

12.4 Enhanced User Interaction

Improving user trust and engagement through natural language interfaces, explanation generation, and visual feedback will be critical for adoption. Transparent communication of system decisions and their implications can bridge the gap between AI-driven control and user understanding.

12.5 Hardware Roadmap and Manufacturing Scale-Up

Immediate priorities include completing Fusion 360 internal-frame models for the Orin Nano form factor and ordering Phase 1 CNC prototypes (5–20 units) for hands-on deployment testing. The transition to Phase 2 manufacturing requires EMC certification (CE/UKCA marking) and thermal validation under the Orin Nano’s 15–40 W variable load envelope. Integration with the Matter protocol will expand device ecosystem compatibility beyond Zigbee, enabling seamless pairing with major smart home brands. Future hardware revisions will explore custom carrier-board designs to replace the Orin Nano developer kit, reducing unit cost and enclosure volume.

12.6 Solar and Battery Integration

A natural extension of the NestShift platform is the integration of rooftop solar photovoltaic (PV) generation and home battery storage. The forecasting and scheduling framework already optimises demand against dynamic grid tariffs; adding solar prediction (cloud-cover forecasting,

panel degradation modelling) and battery state-of-charge management would transform NestShift from a *demand optimiser* into a *net energy manager*. The scheduler would shift deferrable loads to coincide with solar generation peaks, charge the battery during low-price or high-solar periods, and discharge during evening peaks. Early modelling suggests that solar + battery + NestShift optimisation could reduce grid import by 60–80% for south-facing UK properties, cutting annual energy bills by £800–1,200 compared to grid-only operation. This work is reserved for Phase 2 product development.

13 Conclusion

This paper has presented a multi-agent, local-first edge AI architecture for autonomous residential energy optimisation, supported by empirical validation on real smart-meter data, a working neuromorphic learning engine, and a complete hardware-to-software product bridge. By decomposing intelligence into three cooperative agents—energy optimisation, behavioural learning, and lifecycle management—and augmenting this stack with the NARE spiking-neural engine, the system addresses the key limitations of existing HEMS solutions: cloud dependence, privacy risks, poor adaptability, and the gap between theoretical optimisation and deployable products.

The empirical contributions are concrete and reproducible. Trained on **54,197 real half-hourly records** from the Low Carbon London dataset, the LightGBM forecaster achieves $R^2 = 0.89$ on held-out test data with a training time of 0.58 seconds. A dual-head quantile regression architecture (Q90) specifically targets demand spikes, achieving $R^2 = 0.83$ on the upper tail. Applied to genuine load profiles under simulated dynamic tariffs, the greedy scheduler delivers a **26.4% mean cost reduction**—equivalent to **£476/year** per household. These real-data savings substantially exceed the 9–13% observed in complementary digital-twin simulations, underscoring the importance of validating energy optimisation on observed rather than synthetic consumption patterns.

The NARE neuromorphic brain demonstrates a fundamentally different learning paradigm: biologically plausible LIF neurons and STDP synaptic plasticity enable **autonomous habit formation without labelled training data**. In a 30-event teaching scenario, a motion-sensor-to-light synapse potentiated from 0.05 to 0.152, crossing the actuation threshold and enabling autonomous control. This capability is uniquely suited to edge deployment where training data are scarce and privacy is paramount.

Finally, the hardware bridge transforms software results into a tangible product. The custom 6061-T6 aluminium enclosure, NVIDIA Jetson Orin Nano edge platform, and published Bill of Materials define a clear manufacturing path for a bespoke professional installation service. A typical four-bedroom UK installation (hub + sensors + relays + CT clamps + professional fitting) costs £2,000–2,500, yielding a **7.6-year payback** at mean savings (reducing to **3–4 years** with 50% grant funding). Annual software licensing (£180/year) provides recurring revenue and funds continuous model improvement. All software runs locally on the device; no cloud is required after initial OS flash.

Phase 2 commercial deployment targets two engineering hires by Year 1 and 500 UK household installations by Year 2, with expansion into social housing and build-to-rent partnerships. While challenges remain—die-cast tooling capital, long-term model drift, and multi-household generalisation—the evidence presented in this paper establishes a reproducible, economically compelling, and commercially viable proof-of-concept for autonomous residential energy optimisation.

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